PRACTICAL NO.: 1

Experiment Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/ Tools Required: VS Code (text editor), HTML, CSS, JavaScript.

Theory:

JavaScript is a lightweight, high-level, interpreted scripting language used to make web pages interactive and dynamic. It works with HTML and CSS, where HTML provides the structure, CSS provides the design, and JavaScript adds functionality such as displaying messages, validating user input, performing calculations, handling events, and updating webpage content dynamically.

JavaScript can be added to a webpage in three different ways:

1. Inline JavaScript:
Inline JavaScript is written directly inside HTML elements using event attributes such as onclick, onmouseover, etc. It is mainly used for simple tasks and quick demonstrations.

2. Internal JavaScript:
Internal JavaScript is written inside the <script> tag within the HTML document. It is suitable for small web applications where the JavaScript code is limited and only used by a single webpage.

3. External JavaScript:
External JavaScript is written in a separate file with a .js extension and linked to the HTML document using the <script src="filename.js"></script> tag. This method improves code organization, reusability, readability, and maintenance. It is the preferred method for large websites and web applications.

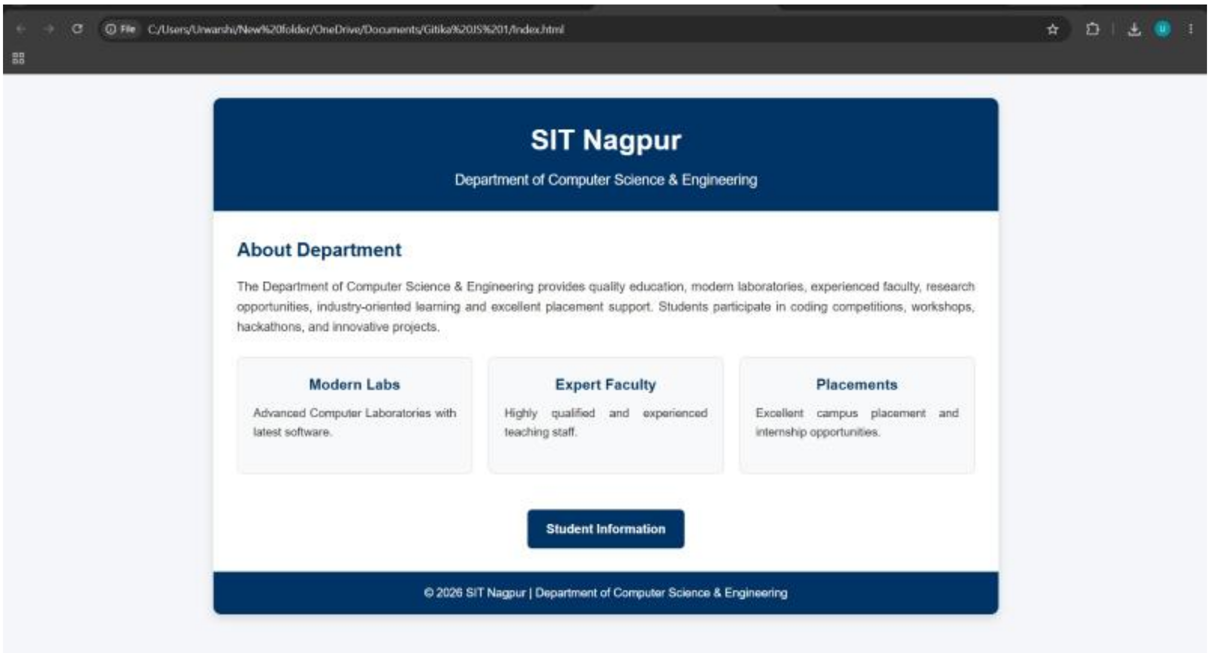
JavaScript also provides several **console methods** that help developers debug and test programs through the browser's Developer Console. Common console methods include:

- **console.log()** – Displays normal messages or variable values.
- **console.info()** – Displays informational messages.
- **console.warn()** – Displays warning messages.
- **console.error()** – Displays error messages.
- **console.table()** – Displays data in a table format.
- **console.clear()** – Clears all console messages.

JavaScript interacts with the **Document Object Model (DOM)** to access, modify, and update HTML elements dynamically. Using DOM methods such as document.getElementById(), document.querySelector(), document.write(), and innerHTML, developers can change webpage content based on user actions.

In this experiment, a webpage is created to demonstrate **inline, internal, and external JavaScript**, use different **console methods**, and display **user information** along with a **welcome message**. This experiment helps in understanding the different methods of including JavaScript in a webpage and the use of browser developer tools for testing and debugging JavaScript programs.

Output:



Result

The experiment was performed successfully. A webpage was created using HTML, CSS, and JavaScript. Inline, internal, and external JavaScript were implemented correctly. User information and a welcome message were displayed successfully, and various console methods produced the expected output in the browser's Developer Console.

Conclusion

The experiment was completed successfully. It demonstrated the three methods of including JavaScript in a webpage: inline, internal, and external. The use of console methods for debugging and displaying messages was also understood. The experiment helped in learning how JavaScript interacts with HTML to create dynamic and interactive webpages, making it an essential technology for modern web development.

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